

Informe de experimento: plan_act_nogate

Generado el 2026-09-18 22:47 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio plan_act_nogate (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: plan_act_nogate
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- System prompt: hash 7da5e2a37ec4 (40 filas)
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-18 22:35
- Costo total: 0.0368 USD (40 items; costo medio 0.00092 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=220122
- Solved global: 95.0% (38/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método plan_act_nogate (system_prompt_hash real de esta fila: 7da5e2a37ec4)

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=plan_act, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
```

- 3. Violations, if any
- 4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

PLAN_SYSTEM_PROMPT (cabecera) −1043 caracteres; primeras 12 líneas de 1424 caracteres; catálogo de tools omitido

You are the planner of a power-system analysis agent. You cannot call tools yourself. Given the user's request, output ONLY a JSON object of the form

```
{
  "plan": [
    {
      "tool": "<tool_name>",
      "args": {
        ...
      }
    },
    ...
  ]
}
```

listing, in execution order, every tool call needed to answer the request. Use only the tools below with exactly these argument names (* = required). Do not include explanations, markdown, or any text outside the JSON.

Available tools:

- load_case(case_name: string*): Load IEEE test case.
- run_powerflow((none)): Run power flow on current case.
- modify_load(bus_id: integer*, p_mw: number*, q_mvar: number): Modify bus load and recompute.
- disconnect_line(from_bus: integer*, to_bus: integer*): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus: integer*, to_bus: integer*): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch((none)): Get most loaded branch.
- run_n1_contingency(top_k: integer, criteria: string, max_candidates: integer): Run N-1 contingency analysis.

...

FINAL_ANSWER_INSTRUCTION (ronda final) −158 caracteres; se añade tras ejecutar el plan

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a non-converged result, say so.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del scoreboard_per_case), en los 4 grupos de metricas_pfagent_definicion: **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta −toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_autonomously_rate (no la columna Solved de arriba, que puede solaparse con Escalated −una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición −resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Escalated	Wrong (silent)	\$
plan_act_nogate	case1440	95.0 (38/40)	0	0.05	95.0 (38/40)	100.0 (40/40)	0	95.0 (38/40)	−	2.4	5503−	0.0 (0/40)	5.0 (2/40)		0.0368

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
plan_act_nogate	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)	80.0 (8/10)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
plan_act_nogate	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)	80.0 (8/10)

Tipos de error de formulación (conteo de items)

Method	ok	wrong_id
plan_act_nogate	38	2

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método `plan_act_nogate`

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
tool_call: load_case({"case_name": "case14"})
tool_call: disconnect_line({"from_bus": 10, "to_bus": 11})
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5930 chars más]
  gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.

1. Summary: The system is operating with some voltage violations.
2. Key numbers: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.06 to 1.09 p.u.
3. Violat ...[207 chars más]
--- Respuesta final ---
The power flow analysis has been completed after disconnecting the line between bus 10 and bus 11.
```

```
1. Summary: The system is operating with some voltage violations.
2. Key numbers: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.06 to 1.09 p.u.
3. Violations: There are multiple overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
4. Suggested next step: Consider implementing voltage control measures or load shedding to address the overvoltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3780;
↳ cost_usd=0.0007; wall=3.9 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-027-s0, case14, difficulty=ambiguous, seed=203647507): Load the IEEE 14-bus system and set the active load at bus
↳ 10 (0-based) to 4.7 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 11, "p_mw": 4.7})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 10, "p_mw": 4.7})
fuente de la traza: archivo de traza completo
--- Ronda 1 [plan] ---
tool_call: load_case({"case_name": "case14"})
tool_call: modify_load({"bus_id": 10, "p_mw": 4.7})
--- Ronda 2 [act] ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.708166484847,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
tool: modify_load({"bus_id": 10, "p_mw": 4.7})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -4.9130978627475494, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm ...[6449 chars más]
  gate: PASS (reasons=-, mismatch=4.77e-09 MW)
--- Ronda 3 [answer] [final] ---
LLM: The IEEE 14-bus system has been successfully loaded and the active load at bus 10 has been set to 4.7 MW.

- Total load: 260.02 MW
- Total generation: 273.56 MW
```

- Total losses: 13.54 MW
- Voltage range: 1.0369 to 1.0900 p.u.

Violations: There are multiple overvoltage violations at buses 1, 6, 7, 8, ...[166 chars más]

--- Respuesta final ---

The IEEE 14-bus system has been successfully loaded and the active load at bus 10 has been set to 4.7 MW.

- Total load: 260.02 MW
- Total generation: 273.56 MW
- Total losses: 13.54 MW
- Voltage range: 1.0369 to 1.0900 p.u.

Violations: There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, 12, and 13.

Suggested next step: Consider running a contingency analysis or generating remedial-action suggestions to address the overvoltage
→ violations.

--- Scoring ---

formulation=wrong_id (exact=False); V_MAE=5.79e-04; F_MAE=1.22; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
→ tokens=4015; cost_usd=0.0007; wall=10.1 s

Ejemplo: con bandera de fallo (claimed/stale/abstained)

request (case14-multistep-022-s0, case14, difficulty=multistep, seed=1083806780): Load case14, disconnect the line between bus 9 and bus 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis
→ ranked by max violations and report the 8 worst outages.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw": 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw": 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})

fuentes de la traza: archivo de traza completo

--- Ronda 1 [plan] ---

tool_call: load_case({"case_name": "case14"})
tool_call: disconnect_line({"from_bus": 9, "to_bus": 10})
tool_call: modify_load({"bus_id": 3, "p_mw": 118.6})
tool_call: modify_load({"bus_id": 4, "p_mw": 53.6})
tool_call: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
--- Ronda 2 [act] ---

tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 264.0857422973543,
→ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 9, "to_bus": 10})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.032452347367612, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0499999999999997, "va_deg": -5.627497640283712, "is_violation": false,
gate: PASS (reasons=-, mismatch=6.74e-09 MW)
tool: modify_load({"bus_id": 3, "p_mw": 118.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000, "va_deg": -5.672176280633835, "is_violation": false,
gate: PASS (reasons=-, mismatch=7.38e-09 MW)
tool: modify_load({"bus_id": 4, "p_mw": 53.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
→ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
→ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000, "va_deg": -5.672176280633835, "is_violation": false,
gate: PASS (reasons=-, mismatch=7.73e-09 MW)
tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
output: {"plot_type": "n1_ranking", "figure_json":
→ "{ \"data\": { \"clonaxis\": false, \"hoverinfo\": \"text\", \"hovertext\": [\"\\u003cb\\u003eOutage\\u003c\\u002fb\\u003e: 10-11
(line)\\u003cbr\\u003e\\u003cb\\u003eConverged\\u003c\\u002fb\\u003e: True\\u003cbr\\u003e\\u003cb\\u003eVoltage violations\\u003c\\u002fb\\u003e: 10-11
[19612 chars más]
gate: n/a (no es un resultado de flujo)

--- Ronda 3 [answer] [final] ---

LLM: The analysis has been completed successfully.

1. Summary: The N-1 contingency analysis identified the worst outages with significant voltage violations.
2. Key numbers: Total load = 264.09 MW, Total generation = 278.14 MW, Total losses = 14.05 MW, Voltage range = 1.0099999999999996 to 1.0900000000000000 ...[311 chars más]
--- Respuesta final ---
The analysis has been completed successfully.

1. Summary: The N-1 contingency analysis identified the worst outages with significant voltage violations.
2. Key numbers: Total load = 264.09 MW, Total generation = 278.14 MW, Total losses = 14.05 MW, Voltage range = 1.0099999999999996 to 1.0900000000000003 pu.
3. Violations: There are multiple voltage violations at several buses, with the worst outage being between bus 10 and bus 11, which has 8 voltage violations.
4. Suggested next step: Consider implementing remedial actions to address the voltage violations, particularly for the worst affected branches.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=stale_state; error=-; tokens=17634;
→ cost_usd=0.0028; wall=5.9 s

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- Todos los métodos que llegan al solver (`plan_act_nogate`) tienen $V_MAE < 1e-4$ p.u. en todos los casos (máximo 1.47e-05 en `plan_act_nogate/case14`).
- El gate verificó 46 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de `plan_act_nogate`: 95.0 % (38/40).
- Formulación exacta agregada sobre métodos con tools: 95.0 % (38/40).
- Formulación por dificultad: mejor `plain` 100.0 % (10/10), peor `ambiguous` 80.0 % (8/10).
- Errores de formulación más frecuentes: `wrong_id=2`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- 1 de 26 items con mutación de red reportaron estado obsoleto (`stale_state`).
- Trazabilidad de números (Faith.) media con tools: 99.6 %.
- Costo total 0.0368 USD; el método más caro fue `plan_act_nogate` con 0.0368 USD (100 %).
- Métodos con tools: 2.35 tool calls medias por item (máximo 5).

6. Archivos

Rutas absolutas bajo `plan_act_nogate.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2]
↳ /power-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard.md
scoreboard_per_case.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p
↳ ower-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard_per_case.json
scoreboard.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p
↳ ower-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p
↳ ower-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_plan_structured/gpt-4o-mini/plan_act_nogate/traces
```